



Large kernel convolution application for land cover change detection of remote sensing images[☆]

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ABSTRACT

In land cover change detection tasks, extracting universal features of changing targets is crucial for achieving precise detection results. A larger receptive field helps the model capture these universal features of changing targets. Although Large Kernel Convolution has been widely used in the field of computer vision, its potential in land cover change detection of remote sensing images has not been fully explored. To address this, a novel Re-parameterization Large kernel Convolution Network for Change Detection (CD-RLKNet) is proposed. CD-RLKNet utilizes Spatial and Temporal Adaptive Fusion Module to preliminarily extract spatial-temporal features from bi-temporal remote sensing images, resulting in a coarse-grained fused feature map. Features Assimilation Assistant Module extracts independent features from land cover information of each temporal, serving as auxiliary information for fine-grained features. Bi-temporal Features Integration Module utilizes large kernel convolution to extract bi-temporal land cover features with a larger receptive field, capturing fine-grained differences in these features. Experiments have been conducted on SYSU-CD, LEVIR-CD and GVLM-CD datasets, and results show that the proposed CD-RLKNet achieves IoU values of 0.6882, 0.8294 and 0.7729, respectively, surpassing the compared SOTA models. The code of CD-RLKNet can be achieved from <https://github.com/juncyan/cdrlnet.git>.

1. Introduction

The essence of multi-temporal land cover change detection (LCCD) is to monitor the process of ground feature changes in the region by using multiple remote sensing images of the same area at different times (Chen et al., 2022b). At present, LCCD technology has a wide range of applications in many fields, such as land use change monitoring (Gao et al., 2024), urban development planning and layout (Wang et al., 2023a; Hu and Ban, 2014), etc (Hu et al., 2024).

Traditional LCCD methods mainly encompass direct image comparison, post-classification comparison (Khan et al., 2020), and strategies based on feature description and image transformation (Mastro et al., 2022; Cheng et al., 2015). Direct image comparison identifies changes by calculating the differences in pixel values between two temporal remote sensing images, with commonly used techniques including image difference, ratio analysis (Todd, 1977), and vegetation index comparison (Singh, 1986). The post-classification comparison approach first independently classifies images from different time periods and then extracts change information by comparing the classification results (Wan

et al., 2019). Additionally, methods based on feature description and image transformation, such as Tasseled Cap transformation and Independent Component Analysis (Bruzzone and Serpico, 1997; Wang et al., 2024a), employ specific mathematical transformation processes to extract feature information reflecting surface changes (Zhao et al., 2017). With the continuous increase in remote sensing and the improvement of resolution, traditional change detection methods have become inadequate to meet the demands for high precision and efficiency (Li et al., 2014; Zhao et al., 2016). Traditional algorithms are mostly unsupervised learning models that heavily rely on empirical design, making them difficult to cope with the complex challenges posed by processing Very High Resolution (VHR) images. Due to traditional LCCD over-reliance on manual experience in design, lack of adaptability and flexibility, resulting in difficulty achieving ideal detection results when processing VHR images (Li et al., 2024), there is an urgent need for efficient and precise processing methods to meet practical application requirements (Ding et al., 2023a). Therefore, developing new LCCD

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techniques suitable for VHR images has become an urgent issue to address.

Deep learning (DL) can learn feature representations from a large amount of data and obtain highly nonlinear models through training, thus enabling efficient processing and analysis of remote sensing images. The emergence of DL has provided an efficient solution for LCCD (Chen et al., 2023). With the development of DL, object-level change detection based on DL has attracted various attentions. Object-based image analysis technology (Wang et al., 2024b) is widely used in high-resolution LCCD tasks. LCCD task has gradually transitioned from object-level to pixel-level.

Convolutional neural networks (CNNs) are one of the commonly used models in DL, lots of CNNs model, such as VGG (Simonyan and Zisserman, 2014), ResNet (He et al., 2016), and MobileNets (Howard et al., 2017), are utilized in features extracting. By training deep CNNs, rich features can be extracted from bi-temporal remote sensing images for the purpose of change detection (Wang et al., 2023b). CNNs-based approaches constitute a holistic end-to-end learning system that operates autonomously, without intermediate manual intervention, ensuring heightened robustness and efficiency throughout the process (Long et al., 2015). From a methodological framework perspective, these methods can be further subdivided into single-input networks and dual-input networks based on image input methods (Jiang et al., 2024). Single-input networks involve cascading bi-temporal images into one image before inputting them into the network (Mascarenhas and Agarwal, 2021). For example, Peng et al. (2019) concatenated bi-temporal image pairs as input for an improved UNet++ network, merging change maps of different semantic levels to generate the final change map. Compared to single-input networks, dual-input networks draw inspiration from siamese networks (Hou et al., 2017), modifying the front-end feature extraction part of CNNs to two structurally identical networks. For instance, Caye Daudt et al. (2018) proposed three fully convolutional neural network frameworks for LCCD, one of which is single-input, while the other two are dual-input. Experimental results indicate that the dual-input network architecture is more suitable for change detection. Zhang et al. (2020) presented a method, which belongs to the dual-input fully convolutional network approach. Bi-temporal images extract deep features through a siamese input network, and during training, direct feedback from the downsampled change map is introduced into the intermediate layers of the network, independently calculating losses to update network parameters. In other hands, attention mechanisms can improve the accuracy of computer vision (Dosovitskiy et al., 2020). In processing high-resolution LCCD, some algorithms combine attention mechanisms with CNNs (Ren et al., 2024b) and have made some progress (Dai et al., 2023).

In LCCD task, obtaining universal feature representations of changing targets under complex backgrounds is crucial for achieving precise detection (Huang et al., 2024). Therefore, the algorithm needs to possess the capability to capture features of large-scale land cover, which requires the model to have a large receptive field (Ren et al., 2024a). In recent years, some research have utilized larger kernels convolution (LKC) to enable models to acquire a larger receptive field. LKC is a convolutional method that extends the size of traditional convolution kernels. By increasing the size of the convolution kernel, LKC can capture large receptive, thereby more effectively extracting features from images (Lau et al., 2024). To further enhance the performance of LKC, Liu et al. proposed ConvNeXt (Liu et al., 2022b) using 7×7 convolution, which is entirely built upon standard CNN, without incorporating Transformer structures, yet it is able to compete with Transformers in terms of accuracy, scalability, and robustness. Furthermore, the authors experimentally demonstrate that altering the convolution kernel size can enhance model performance. RepLNet (Ding et al., 2022b) designs a CNN architecture with large kernels of 31×31 , utilizing re-parameterizable large depth-wise separable convolutions. It provides guidance on how to efficiently design high-performance

CNNs with large kernels. Experimental results show that compared to small kernels, LKC have a larger effective receptive field and higher shape bias. RepLNet is a specific work focused on large-kernel design, which indicates that LKC are more beneficial for downstream tasks in computer vision. Ding et al. (2023b) proposed UniRepLNet, a versatile perception LKC that has demonstrated impressive performance in audio, video, point cloud, time series, and image recognition tasks. UniRepLK employs SE Blocks to increase depth, Dilated Reparam Blocks for convolutional layer reparameterization, and other methods to showcase the robust performance of large kernel convolutions. It also provides strong support for the application of LKC in the field of remote sensing image processing.

In this paper, Re-parameterization Large kernel Convolution Network for Change Detection (CD-RPKNet) is proposed based on Convolution Re-parameterization (Ding et al., 2019). It consists of three modules: Spatial and Temporal Adaptive Fusion Module (STAFM), which initially fuses bi-temporal remote sensing data by extracting temporal and spatial feature differences, forming a coarse-grained fused feature map; Features Assimilation Assistant Module (FAAM), which extracts remote sensing features of each temporal phase as auxiliary features, providing temporal independent features for fine-grained features; and Bi-temporal Features Integration Module (BFIM), which uses Re-parameterization LKC (RLKC) to extract a universal representation of land cover change features from a larger receptive field, ensuring accurate identification of the independence of the detected target and background features. To enhance BFIM's ability to induce biases in remote sensing features, RepLK Information Fusion Block (RepIFB) is designed in BFIM using multi-dilation convolution reparameterization. This not only ensures that the receptive field can obtain sufficient inductive biases for land cover features but also avoids wasting computing power due to excessively large convolution kernels. The innovations of this article are summarized as follows:

- (1) STAFM is proposed to fuse the spatio-temporal feature differences in remote sensing images, initially extracting a coarse-grained feature map of changing targets.
- (2) FAAM is proposed to extract independent features from remote sensing images of different temporal phases as auxiliary features, thus enhancing the differences between features of various temporal.
- (3) BFIM is proposed, which utilizes RLKC to fuse the coarse-grained features from STAFM and the auxiliary features from FAAM, aiming to extract fine-grained features rich in universal representations of changing targets.
- (4) RepIFB is designed, employing reparameterization to avoid computational complexity while maintaining a large receptive field.

2. Methods

2.1. Overview

The overall structure of CD-RLKNet is shown in Fig. 1. CD-RLKNet's feature extraction section consists of three modules: Features Assimilation Assistant Module (FAAM), Spatial and Temporal Adaptive Fusion Module (STAFM) and Bi-temporal Features Integration Module (BFIM). Among them, STAFM and BFIM are used for fine-grained feature extraction from bi-temporal remote sensing images. Initially, STAFM is employed to fuse the bi-temporal land cover data preliminarily, obtaining a coarse-grained fused feature map f_1 for the bi-temporal remote sensing images. Subsequently, BFIM is used to further extract more comprehensive and fine-grained land cover features. Given the distinctive characteristics of LCCD tasks, a large receptive field is essential for comparing the similarities and differences in bi-temporal land cover information. Therefore, in BFIM, we designed the Bi-temporal Features Extraction based on Large Kernel Block (BFE-LKB), which employs RLKC to extract bi-temporal land cover features with a larger

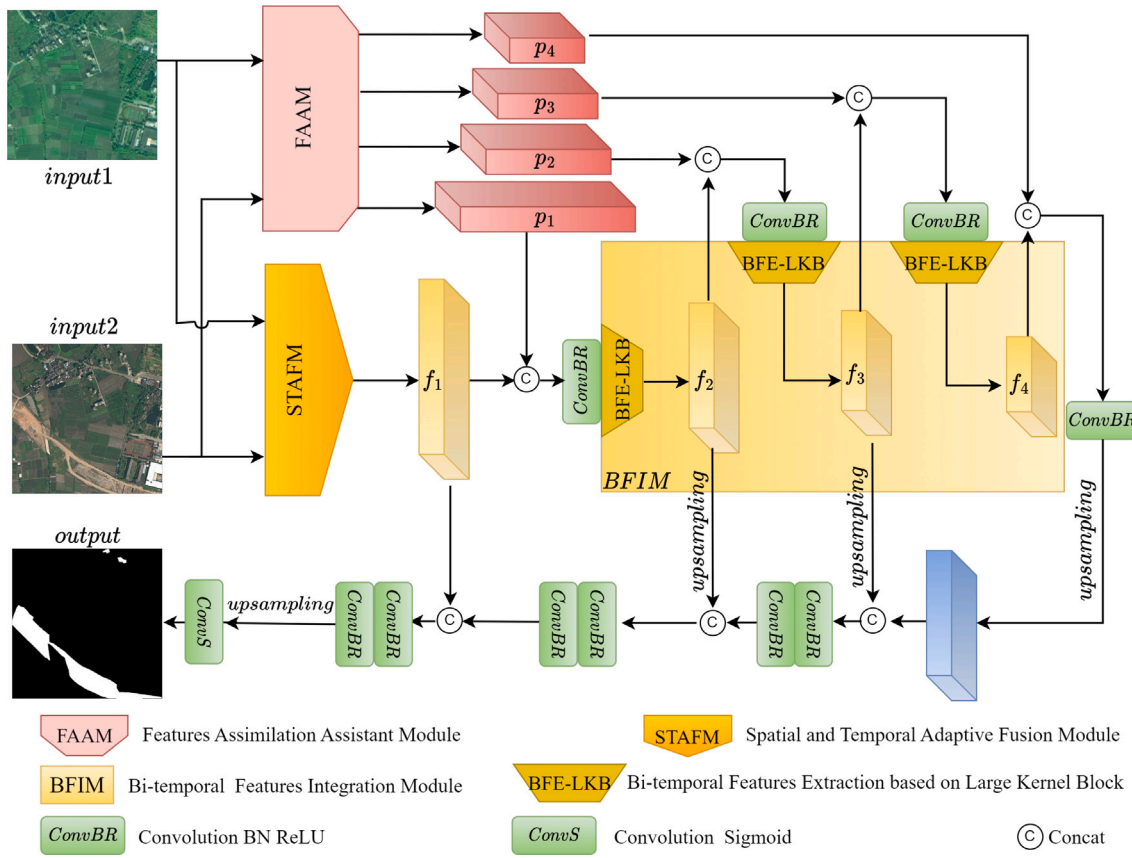


Fig. 1. Architecture of proposed CD-RLKNet, where Features Assimilation Assistant Module (FAAM) and Spatial and Temporal Adaptive Fusion Module (STAFM) are for extracting bi-temporal land cover features. STAFM initially integrates temporal and spatial information from remote sensing images, while FAAM extracts independent feature information from different temporal of remote sensing images and generates auxiliary features accordingly. Bi-temporal Features Integration Module (BFIM) is then utilized to integrate coarse-grained features and auxiliary features to obtain these fine-grained features. To enhance BFIM's ability to extract universal features of changing targets, Bi-temporal Features Extraction based on Large Kernel Block (BFE-LKB) is designed to use RLKC to extract widely dependent features of changing targets from a large receptive field.

receptive field. This allows us to capture the large-scale background features of the detected targets. In Fig. 1, BFIM is composed of three BFE-LKBs and ConvBR. The auxiliary features extracted by FAAM must be compressed through ConvBR before their shape can meet the input requirements of BFE-LKB. The location of ConvBR and BFE-LKB here indicates their computational order.

Considering that the bi-temporal remote sensing images are fused into a single entity after passing through the STAFM, the fine-grained features further extracted by BFIM may lack independence between features from different time phases. To address this issue, we designed the FAAM as an auxiliary module to separately extract land cover features from each time phase, resulting in auxiliary feature maps p_1 , p_2 , p_3 and p_4 . In BFIM, BFE-LKB is used in sequence to extract features fused from f_1 and p_1 , yielding f_2 , f_3 and f_4 . During the feature restoration phase of CD-RLKNet, the features extracted from the backbone network are gradually restored through upsampling to obtain the final prediction results for land cover change information.

2.2. Spatial and temporal adaptive fusion module

LCCD involves the use of remote sensing images and related geospatial data covering the same surface area at different times to identify and analyze changes in the features of that area. To achieve this, in CD-RLKNet, we designed STAFM to extract both temporal and spatial features from bi-temporal remote sensing images. STAFM maps the spatio-temporal features from the original remote sensing images into a coarse-grained fused feature map f_1 for the bi-temporal land cover features.

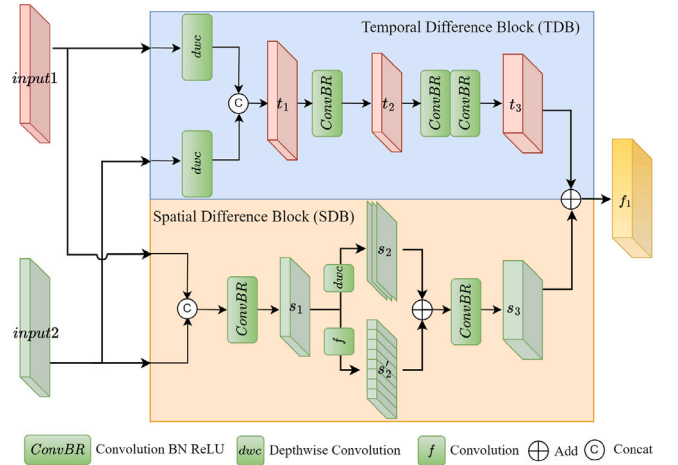


Fig. 2. Architecture of STAFM, which is composed of Temporal Difference Block (TDB) and Temporal Difference Block (SDB). TDB employs a post-fusion approach to extract temporal difference features from remote sensing images, while SDB utilizes a front-fusion method to capture spatial difference features.

The structure of STAFM is shown in Fig. 2. As can be seen from Fig. 2, STAFM is divided into Temporal Difference Block (TDB) and Spatial Difference Block (SDB), which extract temporal and spatial feature differences from the input data, respectively. In the task of LCCD, temporal differences refer to the variations in features over time,

which require capturing the evolution of characteristics at different temporal points. To effectively extract temporal differences, TDB initially employs depth-wise convolution (DWC) with a larger convolution kernel of 7×7 to independently extract input features before concatenating them. This approach allows for a more comprehensive capture of feature changes across temporal points. Spatial differences refer to the variability of features within the same image, such as texture disparities between different types of land cover. For spatial differences, SDB first concatenates input features to highlight the spatial differences of land cover features at different phases, then uses DWC with kernel of 7×7 and a convolution with kernel of 1×1 to integrate block features from different channels and generate spatial difference features. TDB and SDB within STAFM employ distinct computational methods. The rationale behind this is due to the inherent differences between temporal and spatial feature variations in remote sensing images, and the necessity to effectively capture these differences to accomplish the task of change detection.

In TDB, depth-wise convolution (DWC) with kernels of 7×7 is first used to extract input features separately, the shape of $input_1$ and $input_2$ are $3 \times H \times W$, which are then concatenated to obtain t_1 , with shape of $6 \times H \times W$. After that, t_2 is calculated through (1):

$$t_2 = \rho(bn(f_2^{3 \times 3}(t_1))) \quad (1)$$

where ρ indicates ReLU, bn indicates BatchNormalization, $f_2^{3 \times 3}$ indicates convolution with kernels of 3×3 and stride of 2. The same symbols below have the same meaning.

After (1), the size of t_2 being half of t_1 , the shape of t_2 is $6 \times \frac{H}{2} \times \frac{W}{2}$. Subsequently, convolution is used to expand the number of channels in t_2 to obtain t_3 . The calculation formula is as follows:

$$t_3 = \rho(bn(f_2^{3 \times 3}(\rho(bn(f_2^{3 \times 3}(t_2))))) \quad (2)$$

After applying (2), we expand the number of channels in t_2 to 64 to obtain t_3 , with shape of $64 \times \frac{H}{2} \times \frac{W}{2}$.

In SDB, we first concatenate the input features, then compress the feature size to obtain s_1 , with shape of $6 \times \frac{H}{2} \times \frac{W}{2}$. The calculation process as (3).

$$s_1 = \rho(bn(f_2^{3 \times 3}(cat(input_1, input_2)))) \quad (3)$$

where cat indicates concatenate, $f_2^{3 \times 3}$ indicates convolution with kernels of 3×3 . The same symbols below have the same meaning.

Subsequently, a depthwise separable convolution (DWC) with a kernel size of 7×7 is used to integrate block features from different channels to obtain s_2 . Meanwhile, a 1×1 convolution is employed to linearly integrate cross-channel information from s_1 to produce s'_2 . Then, the shape of s_2 and s'_2 are $6 \times \frac{H}{2} \times \frac{W}{2}$, and the feature channels are expanded through convolution to obtain s_3 , with shape of $64 \times \frac{H}{2} \times \frac{W}{2}$. The calculation process is described in (4).

$$s_3 = \rho(bn(f_2^{3 \times 3}(s_2 + s'_2))) \quad (4)$$

Finally, s_3 and t_3 are summed to obtain f_1 , with shape of $64 \times \frac{H}{2} \times \frac{W}{2}$, which represents the output of the STAFM.

2.3. Features assimilation assistant module

The STAFM extracts bi-temporal remote sensing images to obtain coarse-grained fused feature map f_1 , facilitating subsequent fine-grained feature extraction. Given that the essence of change detection algorithms lies in determining the changing regions through bi-temporal images, the fused f_1 from STAFM may lack independent features of each temporal image. To compensate for more original information and details from different temporal phases and provide richer feature information for subsequent high-dimensional features, enabling more accurate change detection, we designed Features Assimilation Assistant Module (FAAM). The structure of FAAM is shown in Fig. 3.

FAAM adopts siamese structure and uses Assimilation Assistant Feature Extraction Block (AAFEB) to sequentially extract features from different temporal phases, which are then summed to produce the output. For the input features, AAFEB first performs downsampling to obtain a_1 , the calculation process is as (5).

$$a_1 = \rho(bn(f_2^{3 \times 3}(input))) \quad (5)$$

Subsequently, DWC and convolution are used to further extract features from a_1 , and the results are summed to produce a_2 , the calculation process is as (6).

$$a_2 = G(bn(dwc(a_1) + f^{1 \times 1}(a_1))) \quad (6)$$

where dwc indicates DWC, G indicates GeLU. The same symbols below have the same meaning.

Following the previous steps, the channel number of a_2 is then expanded, and the output a_2 is obtained through the following calculation.

$$a_3 = \rho(bn(f_2^{3 \times 3}(a_2))) \quad (7)$$

Unlike SDB, the convolution kernel of DWC in AAFEB can be adjusted according to the size of the feature map. Additionally, AAFEB processes input features from remote sensing images captured at different time stages and is a component of FAAM. As an auxiliary feature extraction branch within the overall network, FAAM's role is reflected in the output of AAFEB, which emphasizes the independence of temporal characteristics as auxiliary features. On the other hand, SDB receives concatenated input features from remote sensing images and generates spatial difference features that represent changes in land cover.

After computation through the FAAM, the auxiliary features p_1 , p_2 , p_3 and p_4 are obtained. The feature channel number are 64, 128, 256, and 512, respectively. The size of p_1 is half of the input feature size, while the sizes of p_2 , p_3 and p_4 are successively half of the previous feature map's size. FAAM independently extracts deep features from images captured at different times and then pairs the corresponding elements from these temporal feature maps. FAAM can complement the independent features from remote sensing images captured at different temporal.

2.4. Bi-temporal features integration module

After using STAFM to extract input features, the resulting f_1 only contains low-level information from bi-temporal remote sensing images. To obtain more complex, advanced, and detailed land cover feature representations, and to further predict the details and texture information in the image during the feature recovery phase, the Bi-temporal Features Integration Module (BFIM) is designed. The structure of FAAM is shown in Fig. 4.

To ensure that BFIM can fully extract detailed contextual features from bi-temporal land cover features, f_1 is concatenated with p_1 as the input feature map for BFIM. Then, Bi-temporal Features Extraction based on Large Kernel Block (BFE-LKB) is used for feature extraction to obtain f_2 . In BFE-LKB, RepLK Information Fusion Block (RepIFB) is designed to extract features from a large receptive field, enhancing the performance of BFE-LKB. The computation process of BFE-LKB is divided into two steps: Feature Extraction and Channels Expansion. Firstly, f_n and p_n , $n = 1, 2, 3, 4$ are concatenated as the input for BFE-LKB, and then a convolution is used to compress the feature channels to obtain b_1 , which is the input for Feature Extraction. The computation as (8).

$$b_1 = \rho(bn(f_2^{3 \times 3}(cat(f_1, p_1)))) \quad (8)$$

The shape of b_1 is the same as both f_1 and p_1 . Feature Extraction adopts a residual structure. It first computes b_1 using convolution to obtain e_1 .

$$e_1 = \rho(bn(f_2^{3 \times 3}(b_1))) \quad (9)$$

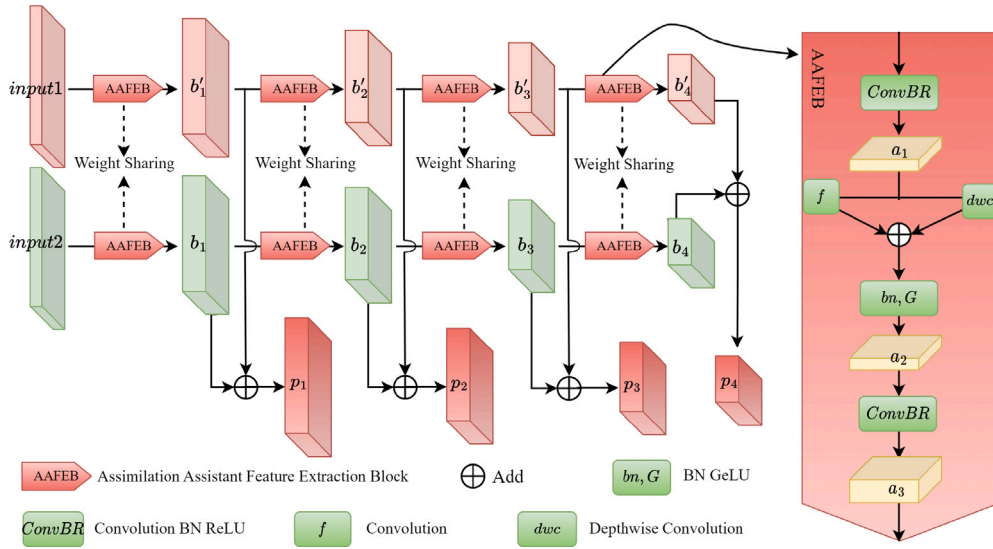


Fig. 3. Architecture of FAAM, which employs an Assimilation Assistant Feature Extraction Block (AAFEb) to extract distinct dimensional feature information from remote sensing images across varying temporal, further integrating these features to preserve their independence from different temporal, thereby mitigating potential interference.

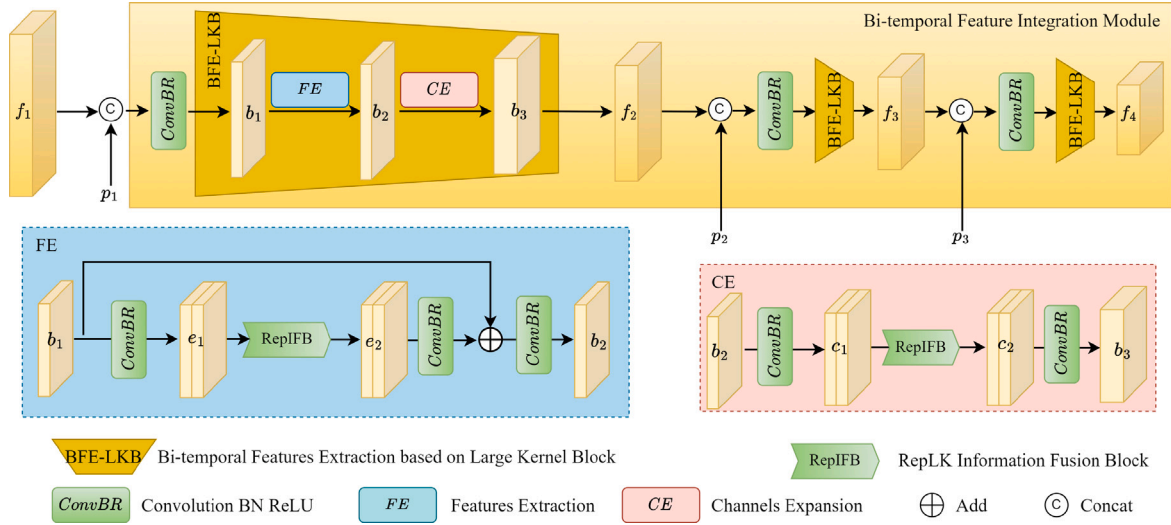


Fig. 4. Architecture of BFIM, where Bi-temporal Features Extraction based on Large Kernel Block (BFE-LKB) is designed by integrating the coarse-grained features generated by FAAM and STAFM, thereby obtaining fine-grained features from bi-temporal land cover features. BFE-LKB initially utilizes Feature Extraction to extract fine-grained features from remote sensing images. Subsequently, Channels Expansion is used to expand the feature channels, aggregating high-dimensional remote sensing features.

Then, a RLKC is used to further extract features, resulting in e_2 .

$$e_2 = Rlkb(e_1) \quad (10)$$

where $Rlkb$ indicates RepIFB. The same symbols below have the same meaning. Afterwards, the shape of e_2 is transformed to be the same as b_1 . Then, b_1 and e_2 are added together. Through calculation (11), the output of Feature Extraction, denoted as b_2 , is obtained. b_2 serves as the input for Channels Expansion.

$$b_2 = \rho(bn(f^{3 \times 3}(\rho(bn(f^{3 \times 3}(e_2))) + b_1))) \quad (11)$$

The role of Channels Expansion is to expand the feature channels. Firstly, for the input b_2 , downsampling is performed to compress its size to half of the original, calculation as (12),

$$c_1 = \rho(bn(f_2^{3 \times 3}(b_2))) \quad (12)$$

Afterwards, RepIFB is used to calculate and obtain c_2 . Finally, a convolution is used to expand channels number of c_2 , resulting in b_3 . The calculation as (13).

$$b_3 = \rho(bn(f^{3 \times 3}(c_2))) \quad (13)$$

b_3 is the output of BFE-LKB, with twice channels number of b_1 and half the size of b_1 . After obtaining f_2 , we continue to concatenate f_2 and p_2 , and repeat the aforementioned process to get f_3 , f_4 . The shapes of f_n and p_n , $n = 1, 2, 3, 4$ are the same.

In LCCD tasks, capturing fine-grained contextual correlations requires maximizing the extraction of differential information between background and changed targets. This necessitates a feature extraction module with a large receptive field. The most direct way to increase the receptive field is to enlarge the convolution kernel size. However, increasing the convolution kernel size leads to a nonlinear increase in model parameters. Too many parameters can easily cause overfitting. Therefore, in BFE-LKB, RepIFB is designed, the structure of RepIFB is shown in Fig. 5.

The structure of RepIFB is simple. Its input features are processed through two distinct paths: one employs standard convolutional, while the other utilizes Depthwise Convolution Reparameterized to Large Kernel Convolution (DWC_RepLK). DWC_RepLK applies DWC with varying dilation rates to extract features from different scales. Through reparameterization, multiple dilated convolutions are combined into a

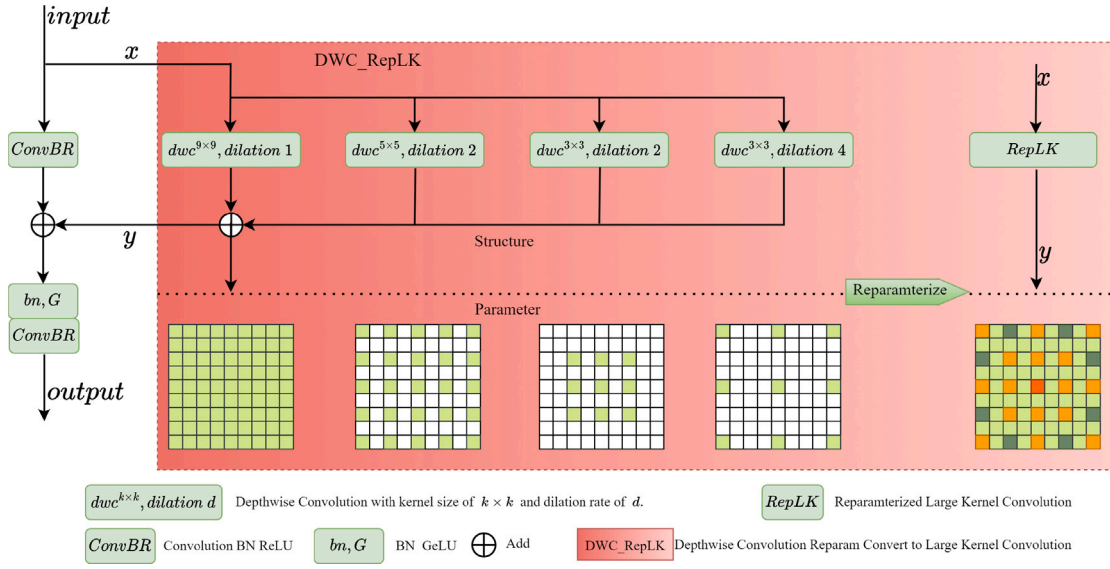


Fig. 5. Architecture of RepIFB. DWC_ReplK obtains a RLKC $RepLK$ by reparameterizing four DWC with different dilation rates. Here, a 9×9 kernel is used as an example, but the actual dilation rates may vary depending on the size of $RepLK$ kernel.

single, larger kernel convolution. The receptive field of convolution is related to the size of the convolution kernel and the dilation rate, and the correspondence is as (14).

$$k_{rf} = (k - 1) \times r + 1 \quad (14)$$

where k_{rf} indicates receptive field of CNNs, k indicates kernel of convolution, r indicates dilation rate of CNNs.

In DWC_ReplK, In DWC_ReplK, four DWC are used. Taking 9×9 as an example, the convolution kernel sizes and dilation rates of each DWC are (9, 1), (5, 2), (3, 2) and (3, 4), respectively. For RLKC, different dilation rates will be adjusted for fusion. Finally, the four DWC will be reparameterized to obtain RepLK. This approach ensures that the receptive field of RepIFB can capture sufficient inductive biases of land cover features while avoiding issues such as wasted computational resources and the introduction of noisy features caused by excessively large convolution kernels. By utilizing these approaches, fine-grained contextual correlations in remote sensing images can be effectively extracted while maintaining model complexity and generalization ability, thereby improving the accuracy of change detection tasks.

3. Experiments and results

3.1. Datasets selected

Extensive experiments were conducted on SYSU-CD (Shi et al., 2021), LEVIR-CD (Chen and Shi, 2020a), GVLM-CD (Zhang et al., 2023) and to verify the practical performance of the proposed CD-RLKNet.

SYSU-CD (Shi et al., 2021) focuses on RGB bands and collects land cover change data from Hong Kong spanning from 2007 to 2014. In addition, compared to other datasets, SYSU-CD offers complementary examples regarding the evolution of high-rise buildings and ports. However, due to the inclusion of a broader range of change types, SYSU-CD contains more distracting information, which making detection of change targets more challenging.

LEVIR-CD (Chen and Shi, 2020a) consists of RGB spectral data capturing building change detection in various cities in Texas from 2002 to 2018. These cities include Austin, Lakeway, Bee Cave, Buda, Kyle, Manor, Pflugerville, and Dripping Springs. LEVIR-CD dataset specifically designed for change detection in remote sensing imagery. It comprises 637 high-resolution (0.5 meters per pixel) Google Earth image pairs, with each image pair measuring 1024×1024 pixels. It

encompasses a wide range of building types, including villas, high-rise apartments, small garages, and large warehouses, among others. Additionally, it contains 31,333 individual instances of change, making it highly suitable for training and testing LCCD algorithms. Its high-resolution imagery and captures significant land use changes over extended time spans.

GVLM-CD (Zhang et al., 2023) is the first large-scale open source VHR landslide mapping dataset. These data collect change information from 2010 to 2021, specifically capturing land slide data before and after natural disasters in various locations: A Luoi, Asakura, Askja, Big Sur, Chimanamani, Jiuzhaigou, Kaikoura, Kodagu, Kupang, Kurucasile, Los Lagos, Osh, Santa Catarina, Shimen, Taitung, Tbilisi, and Tenejapa. It includes 17 pairs of bi-temporal VHR RGB images at a resolution of 0.59 meters obtained through the Google Earth service. These images were cropped to a size of 256×256 without any overlap, resulting in a total of 7327 pairs. GVLM can be used to develop and evaluate machine/deep learning models for change detection, etc.

3.2. Experimental setting and metrics

All experiments in this chapter were trained and tested on an NVIDIA GeForce RTX 4090 using the Adam optimizer. The learning rate was dynamically adjusted using the Poly learning rate scheduling strategy. The experiment adopts CrossEntropy to calculate the loss function. The Poly learning rate scheduling formula is as (15).

$$lr = initial_lr \times (1 - \frac{iter}{iters_num})^{power} \quad (15)$$

where $initial_lr$ refers to the initial learning rate. $iter$ represents the current iteration count. $iters_num$ denotes the total number of iterations. $power$ is a constant, often set to 0.9, which controls the speed of learning rate decay.

For intuitive comparison of the performance of models, this paper selects kappa, IoU, PA, F1 to quantitatively compare the performance of each model on different datasets. In which, Kappa measures the agreement between the predicted and actual labels beyond what would be expected by chance, IoU measures the overlap between the predicted and actual changed areas, PA calculates the proportion of correctly classified pixels out of all pixels, F1 is a weighted average of precision and recall, it is particularly useful when there is an uneven class distribution. The calculation formulas for each metric are as follows.

$$PA = \frac{TP + TN}{TP + TN + FN + FP} \quad (16)$$

$$IoU = \frac{TP}{TP + FP + FN} \quad (17)$$

$$Precision = \frac{TP}{TP + FP} \quad (18)$$

$$Recall = \frac{TP}{TP + FN} \quad (19)$$

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (20)$$

$$Pe = \frac{(TP + FP) \times (TP + FN) + (TN + FN) \times (TN + TP)}{(TP + TN + FP + FN)^2} \quad (21)$$

$$Kappa = \frac{PA - Pe}{1 - Pe} \quad (22)$$

where TP is true positive, which represents number of pixels correctly identified as changed; TN is true negative, which represents number of pixels correctly identified as unchanged; FN is false negative, which represents number of pixels incorrectly identified as unchanged when they have indeed changed; FP is false positive, which represents number of pixels incorrectly identified as changed when they have not. The same symbols below have the same meaning.

3.3. Comparison with state-of-the-art methods

To comprehensively evaluate the performance of CD-RLKNet, we selected SOTA models for comparison. Based on the methods used by these models, they can be categorized into CNN-based models and Transformer-based models. It should be noted that both the attention mechanism and Transformer utilize the concept of attention to enhance the representational capability of the model. When classifying models, those based on the attention mechanism and those that combine CNN with the attention mechanism are categorized as Transformer-based models.

Among them, CNN-based models as follows: **FC-siam-diff** (Caye Daudt et al., 2018) is the first change detection model that utilizes FCN and Siamese structures to share weights, effectively capturing change information by directly comparing or fusing image features. **DTCDSN** (Liu et al., 2020) combines residual networks with feature pyramid networks, to effectively extract multi-scale features, which helps to accurately identify areas of change. **DSIFNet** (Zhang et al., 2020) employs convolution two-stream architecture to extract highly representative deep features from bi-temporal images. Subsequently, the multi-level deep features of the original image are fused with the image difference features through an attention module for change map reconstruction. **SNUNet** (Fang et al., 2022) mitigates the loss of positioning information in deep networks through a tight information transmission mechanism, introducing a feature supervision mechanism to enhance the representation ability of features at different semantic levels. **RDPNet** (Chen et al., 2022a) introduces edge loss to increase the penalty on detail areas, enhancing the network's focus on boundaries and small regions. **BiSRNet** (Ding et al., 2022a) enhances the accuracy of change detection by designing a dual-temporal semantic reasoning module that models both single-temporal and cross-temporal semantic correlations. This design enables the network to better understand the changes in image objects at different time points, thereby improving the accuracy of change detection. **FCCDN** (Chen et al., 2022b) improves the accuracy of high-resolution image change detection through feature constraints and attention mechanisms. **USSFCNet** (Lei et al., 2023) employs multi-scale decoupling convolution to flexibly capture the multi-scale features of changing objects. **CGNet** (Han et al., 2023) uses a variational guidance module to capture long-range dependencies, addressing the field limitation issues of CNNs.

Transformer-based models as follows: **STANet** (Chen and Shi, 2020b) employs a self-attention mechanism to effectively model spatio-temporal relationships. By partitioning the changing features across multiple scales, it captures spatio-temporal dependencies at

different scales, enhancing the representational capacity for objects of varying sizes. **BIT-CD** (Hao Chen and Shi, 2021) enhances the original features by using context-rich token feedback, improving model efficiency and stability. **DSAMNet** (Shi et al., 2021) employs the concept of Transformer to introduce a convolutional block attention module in change detection tasks, offering more discriminative features and enhancing model accuracy. **ChangeFormer** (Bandara and Patel, 2022) employs a complete Transformer structure to efficiently render multi-scale long-range details. **MSCANet** (Liu et al., 2022a) combines CNN and Transformer to enhance feature representation using multi-scale context aggregation modules. **ICIFNet** (Feng et al., 2022) facilitates feature interaction between CNN and Transformer through a convolution attention module. Two cross-scale fusion schemes based on attention are introduced to strengthen the integration of information at different resolutions.

These models cover the evolution from CNN to Transformer, showcasing the technological development trajectory in the field of remote sensing image change detection. Through comparison, the advantages of CD-RLKNet in remote sensing change inspection can be highlighted. The experimental results are presented in Table 1. In Type column of Table 1 C indicates CNN-based models, T indicates Transformer-based models. As shown in Table 1, on SYSU-CD, CD-RLKNet's Kappa is 2.95% higher than the second RSPNet. This indicates that the prediction results of CD-RLKNet have a high consistency with the actual classification results, meaning that the model can accurately identify and classify different land features or changes. CD-RLKNet's IoU is 2.93% higher than the second BiSRNet. This indicates that the predicted results of CD-RLKNet and the actual change areas have a large overlap, demonstrating the model's accuracy in locating change areas. CD-RLKNet's PA is 1.03% higher than the second STANet. This indicates that CD-RLKNet can correctly classify most pixels, demonstrating the model's accuracy in overall change detection tasks. CD-RLKNet's F1 is 1.06% higher than the second BiSRNet. This indicates that CD-RLKNet can accurately identify changing regions while also covering most of the actual changing areas, which means the model maintains good performance in handling various complex scenarios. Likewise, the performance of CD-RLKNet on LEVIR-CD and GVLN-CD also surpasses that of the comparison models. On LEVIR-CD, CD-RLKNet's Kappa is 0.19% higher than the second-best CGNet; CD-RLKNet's IoU is 0.67% higher than the second BiSRNet; CD-RLKNet's PA is higher by 0.02% than the second CGNet; and CD-RLKNet's F1 is 0.99% higher than the second BiSRNet. On GVLN-CD, CD-RLKNet's Kappa is 2.00% higher than the second MSCANet; CD-RLKNet's IoU is higher by 3.01% than the second ICIFNet; CD-RLKNet's PA is 0.18% higher than the second RDPNet; and CD-RLKNet's F1 is higher by 1.41% than the second RDPNet. STAFM combines temporal features and spatial features of remote sensing images, effectively capturing and integrating the temporal and spatial difference information in bi-temporal remote sensing images, generating a coarse-grained fused feature map. This provides high-quality preliminary information for subsequent fine-grained feature extraction by BFIM and FAAM. STAFM ensures that CD-RLKNet can accurately identify and analyze the details of land cover changes in bi-temporal remote sensing images, thereby achieving high-precision change detection tasks. The above data analysis fully demonstrates the excellent performance of CD-RLKNet in LCCD tasks, and experimental results show that STAFM can enhance the model's ability to handle complex tasks.

To facilitate an intuitive comparison of the prediction results of each model, this section visualizes the prediction results of various models. The visualization is presented in Fig. 6. It should be noted that due to space limitations, only the prediction results of some models with better performance metrics are displayed. For a more intuitive comparison of model results, the prediction results of each model are colored based on TP, TN, FN and FP. White represents TP, black represents TN, maroon represents FN, green represents FP. In the prediction results, a lower proportion of maroon and green indicates better model performance.

Table 1

Comparisons with existing networks, where *C* indicates CNN-based models, *T* indicates Transformer-based models. The best scores are marked in bold.

Type	Methods	SYSU-CD				LEVIR-CD				GVLM-CD			
		Kappa	IoU	PA	F1	Kappa	IoU	PA	F1	Kappa	IoU	PA	F1
<i>C</i>	FC-Siam-diff	0.6001	0.6161	0.8823	0.7208	0.8742	0.7846	0.9902	0.7688	0.7988	0.7246	0.9789	0.8123
	DTCDSCN	0.5929	0.6074	0.8718	0.7454	0.8261	0.7164	0.9835	0.8913	0.7808	0.6567	0.9774	0.8047
	DSIFN	0.6607	0.6370	0.8893	0.7702	0.6190	0.7584	0.9782	0.8087	0.7527	0.7102	0.9703	0.8116
	SUNet	0.6386	0.6178	0.8823	0.7629	0.8421	0.7382	0.9859	0.8362	0.8188	0.7334	0.9766	0.8376
	RDPNet	0.6951	0.6552	0.8979	0.8014	0.8568	0.7585	0.9888	0.9082	0.8295	0.7265	0.9832	0.8421
	BiSRNet	0.6951	0.6589	0.8982	0.8033	0.8985	0.8227	0.9919	0.9112	0.8131	0.7313	0.9816	0.8321
	FCCDN	0.6806	0.6526	0.8907	0.7891	0.8900	0.8091	0.9913	0.9105	0.7656	0.7364	0.9768	0.8418
	USSFCNet	0.6421	0.6227	0.8820	0.7982	0.8469	0.7455	0.9861	0.8923	0.7533	0.7191	0.9776	0.8165
	CGNet	0.6852	0.6003	0.8966	0.7991	0.8997	0.8215	0.9920	0.9103	0.8211	0.7104	0.9819	0.8307
<i>T</i>	STANet	0.6851	0.6140	0.8984	0.8072	0.7833	0.6526	0.9850	0.8915	0.8109	0.7291	0.9806	0.8109
	BIT-CD	0.6456	0.6636	0.8853	0.7803	0.8928	0.8136	0.9916	0.8974	0.8126	0.7146	0.9815	0.8107
	DSAMNet	0.5309	0.6312	0.8200	0.7648	0.8499	0.7849	0.9876	0.8914	0.7954	0.7347	0.9775	0.8103
	ChangeFormer	0.6162	0.6089	0.8729	0.7325	0.8468	0.8211	0.9862	0.9013	0.8182	0.7066	0.9814	0.8280
	MSCANet	0.5740	0.6168	0.8686	0.7917	0.8029	0.6837	0.9823	0.9014	0.8354	0.6449	0.9583	0.8419
	ICIFNet	0.6644	0.6153	0.8886	0.8003	0.8986	0.8051	0.9907	0.9070	0.7707	0.7428	0.9746	0.8358
Ours		0.7246	0.6882	0.9089	0.8139	0.9016	0.8294	0.9922	0.9211	0.8524	0.7729	0.9850	0.8562

The first to third rows in Fig. 6 represent the prediction results on the SYSU-CD test set, the fourth to sixth rows show the prediction results on the LEVIR-CD test set, and the seventh to ninth rows display the prediction results on the GVLM-CD test set. The first to third columns represent the input data and groundtruth, respectively. The fourth to tenth columns display the prediction results of BIT-CD, ChangeFormer, DSAMNet, USSFCNet, RDPNet, BiSRNet and Ours, respectively.

In SYSU-CD, the relatively large green and red areas in the prediction results of various models indicate a significant number of false positives and false negatives. This is due to the diverse change types and complex backgrounds of changing targets in SYSU-CD dataset. For instance, the first row shows forest changes against a background of ocean and land, the second row depicts road changes against a backdrop of forested areas, and the third row features building changes set against a forested background. Thus, SYSU-CD poses a considerable challenge. CD-RLKNet employs a super large kernel convolution, enabling it to capture features with a larger receptive field, effectively distinguishing the background information of changing targets and extracting their universal features. Consequently, CD-RLKNet's predictions closely align with the labels.

LEVIR-CD is a dataset focused on building change detection. Compared to SYSU-CD, LEVIR-CD has a singular detection target, typically comprising regular geometric shapes. However, the changing targets are complex, and there are significant differences between remote sensing data across different time periods, resulting in notable spatio-temporal feature disparities, as evident in the fourth and sixth rows. This leads to a considerable amount of false detections. STAFM in CD-RLKNet adeptly distinguishes the spatial-temporal disparities in remote sensing imagery, providing an initial fusion of features in changing regions. STAFM substantially enhances CD-RLKNet's performance on the LEVIR-CD dataset.

Regarding GVLM-CD, landslide edges are highly irregular and jagged. Therefore, extracting universal edge features of landslide targets is crucial. CD-RLKNet utilizes BFIM to extract these universal edge features on a large scale, while FAAM captures independent features from GVLM-CD data across different time periods. This effectively supplements independent temporal information for fine-grained features, enabling precise differentiation of landslide edges.

4. Discussion

4.1. Modules ablation study

To validate the effectiveness of each module, ablation studies were conducted in this section. The results of the ablation experiments are presented in Table 2.

In Table 2, “FAAM out” represents the experiment where FAAM is not used, which means the auxiliary branch is not applied. “STAFM out” indicates that module STAFM is not used, instead, the input features are directly concatenated and processed using a ResBlock. Lastly, “BFIM out” signifies that module BFIM is not utilized. “ResNet”, “RepLK31B”, “ViT” and “SwinT” represent experiments where BFIM has been replaced by ResNet50 (He et al., 2016), RepLK31B (Ding et al., 2022b), ViT (Dosovitskiy et al., 2020) and SwinTransformer (Liu et al., 2021), respectively, for the purpose of extracting fine-grained dual-temporal remote sensing features.

Based on the data in Table 2, it can be observed that the BFIM has the greatest impact on model performance. When the BFIM is not used, the IoU on the validation sets of SYSU-CD, LEVIR-CD, and GVLM-CD are lower by 5.41%, 2.67%, and 3.77%, respectively, compared to the standard model. Utilizing ResNet50, RepLK31B, ViT and SwinTransformer to replace BFIM resulted in lower performance metrics across the board compared to the standard model. This data fully demonstrates that BFIM, utilizing RepIFB, has the capability to capture contextual information related to changing targets from a larger receptive field and represent universal features in complex backgrounds. It can efficiently handle change detection problems in bi-temporal remote sensing images. Although the impact of FAAM and STAFM on CD-RLKNet performance is not as significant as that of BFIM, these two modules can still enhance the recognition performance of the model in change detection tasks from different perspectives.

4.2. Kernel size of large kernel convolution ablation study

To verify the impact of convolution kernel size on change detection results, this section gradually increases the kernel size starting from 9. Table 3 presents the metrics of CD-RLKNet with different convolution kernel sizes. As shown in the table, the convolution kernel size is positively correlated with the number of model parameters. However, as the convolution kernel size increases, the rate of IoU improvement flattens out. Consequently, we chose a convolution kernel size of 33 for CD-RLKNet.

To compare the effect of convolution kernel size on the model, Fig. 7 shows the Grad-CAM visualization of CD-RLKNet with different convolution kernels. It can be observed that as the convolution kernel size increases, the model assigns greater weights to the pixels representing changing regional features.

To further compare the performance of RepIFB, experiments are conducted with its replacement by RLKC blocks from ConNeXt(ConX) (Liu et al., 2022b), RepLK (Ding et al., 2022b), Lark (Ding et al., 2023b). The experimental results are presented in Table 4, where kernels with ConX are 7×7 , kernels with RepLK and Lark are 33×33 ,

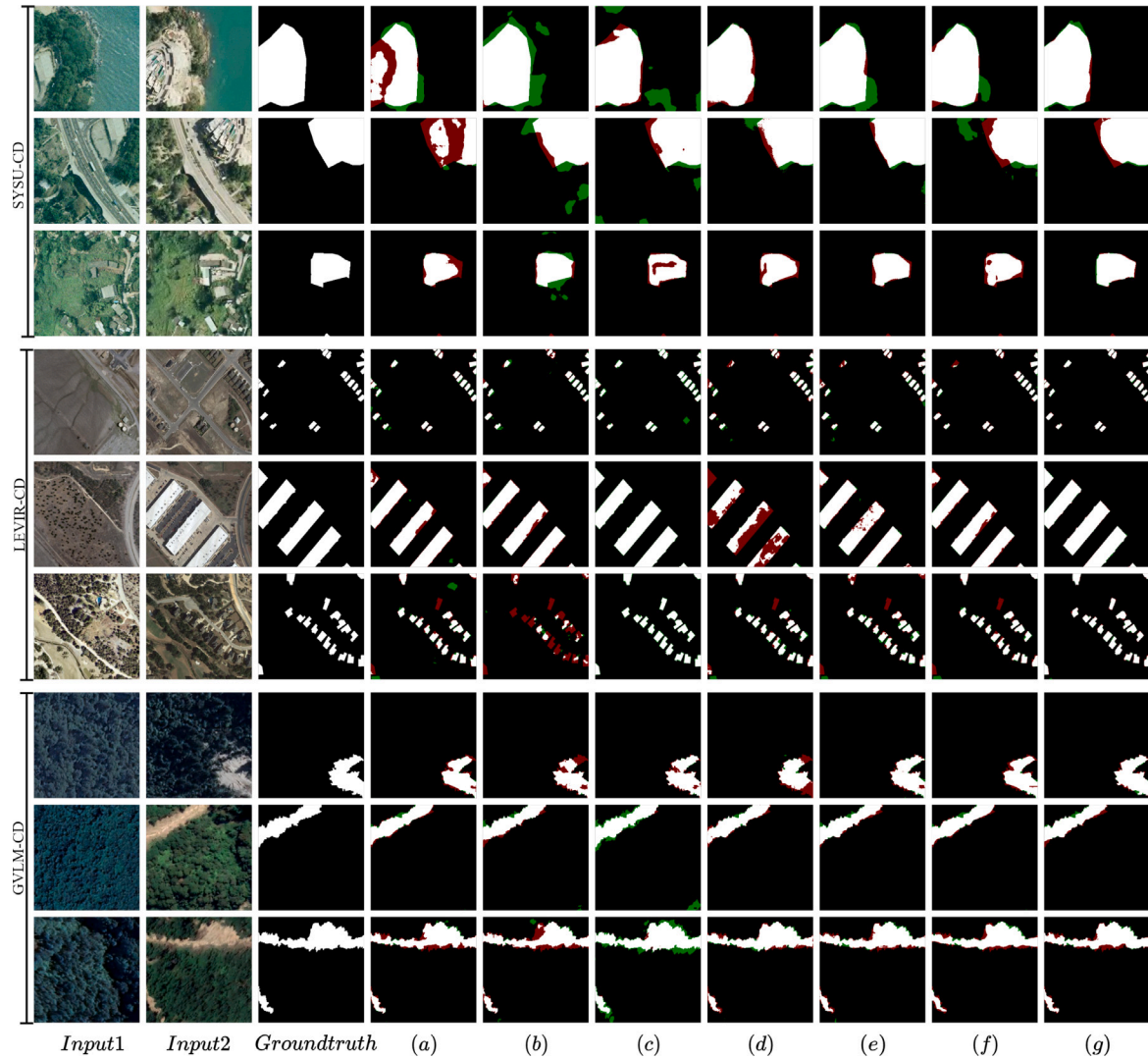


Fig. 6. Visualization and qualitative comparisons. The first to third rows show examples from SYSU-CD, the fourth to sixth rows display data from LEVIR-CD, and the seventh to ninth rows exhibit data from GVLM-CD. The first to third columns represent remote sensing images at different time along with the corresponding ground truth. In the Groundtruth, black represents unchanged regions, while white indicates changed regions. From the fourth row to the tenth row show the visualization results of different models: (a) BIT-CD (Hao Chen and Shi, 2021), (b) ChangeFormer (Bandara and Patel, 2022), (c) DSAMNet (Shi et al., 2021), (d) USSFCNet (Lei et al., 2023), (e) RDPNet (Chen et al., 2022a), (f) BiSRNet (Ding et al., 2022a), and (g) CD-RLKNet. In prediction results, white represents TP, black is TN, maroon indicates FN, and green stands as FP. In short, the lower the proportion of maroon and green, the better the predictive performance of the model. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 2

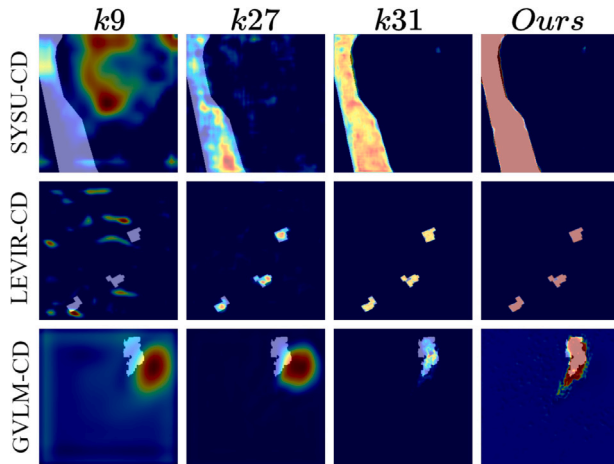
Ablation study of modules. The best scores are marked in bold.

Datasets	Metric	FAAM out	STAFM out	BFIM out	ResNet	RepLK31B	ViT	SwinT	Ours
SYSU-CD	Kappa	0.7077	0.7115	0.6778	0.7038	0.7132	0.6998	0.6931	0.7246
	IoU	0.6595	0.6583	0.6341	0.6558	0.6557	0.6371	0.6273	0.6882
	PA	0.9025	0.9058	0.8921	0.9004	0.9050	0.9012	0.8986	0.9089
	F1	0.7903	0.7896	0.7961	0.8070	0.8032	0.8007	0.8013	0.8139
LEVIR-CD	Kappa	0.8877	0.8905	0.8840	0.8970	0.8916	0.8963	0.8889	0.9016
	IoU	0.8101	0.8099	0.8027	0.8119	0.8067	0.8192	0.8078	0.8294
	PA	0.9909	0.9912	0.9909	0.9918	0.9912	0.9917	0.9909	0.9922
	F1	0.9086	0.9105	0.9073	0.9105	0.9131	0.9153	0.9147	0.9211
GVLM-CD	Kappa	0.8434	0.8473	0.8429	0.8426	0.8351	0.8437	0.8117	0.8524
	IoU	0.7529	0.7597	0.7352	0.7417	0.7510	0.7624	0.7483	0.7729
	PA	0.9842	0.9841	0.9837	0.9841	0.9831	0.9841	0.9812	0.9850
	F1	0.8197	0.8376	0.8157	0.8039	0.8055	0.8506	0.8518	0.8562

Table 3

Ablation study of using different kernel sizes. The best scores are marked in bold.

Datasets	Metric	9	13	15	17	21	25	29	31	33 (Ours)
SYSU-CD	Kappa	0.7150	0.7169	0.7182	0.7192	0.7200	0.7221	0.7238	0.7243	0.7246
	IoU	0.6347	0.6575	0.6632	0.6656	0.6674	0.6683	0.6689	0.6849	0.6882
	PA	0.9059	0.9078	0.9082	0.9065	0.9075	0.9077	0.9074	0.9089	0.9089
	F1	0.7536	0.7765	0.7905	0.7976	0.8092	0.8116	0.8133	0.8138	0.8139
LEVIR-CD	Kappa	0.6561	0.8884	0.8930	0.8937	0.8957	0.8962	0.8978	0.8982	0.9016
	IoU	0.8164	0.8172	0.8182	0.8203	0.8223	0.8224	0.8247	0.8249	0.8294
	PA	0.9620	0.9912	0.9916	0.9915	0.9916	0.9918	0.9918	0.9919	0.9922
	F1	0.8726	0.9100	0.9165	0.9195	0.9197	0.9203	0.9209	0.9209	0.9211
GVLM-CD	Kappa	0.8388	0.8409	0.8417	0.8462	0.8476	0.8486	0.8522	0.8524	0.8524
	IoU	0.7304	0.7550	0.7669	0.7679	0.7681	0.7693	0.7711	0.7717	0.7729
	PA	0.9832	0.9836	0.9834	0.9841	0.9841	0.9844	0.9846	0.9850	0.9850
	F1	0.8307	0.8419	0.8479	0.8492	0.8534	0.8541	0.8543	0.8558	0.8562

**Fig. 7.** Grad-CAM of different convolution kernel models. Here, k_9 , k_{27} and k_{31} represent RepIFB convolution kernels of sizes 9, 27 and 31, respectively. *Ours* represents kernel of sizes 33.**Table 4**

Ablation study of using different LKC blocks. The best scores are marked in bold.

Dataset	Blocks	Kappa	IoU	PA	F1
SYSU-CD	ConX	0.7184	0.6669	0.9073	0.8591
	RepLK	0.7222	0.6737	0.9077	0.8611
	Lark	0.7222	0.6877	0.9097	0.8610
	Ours	0.7246	0.6882	0.9089	0.8623
LEVIR-CD	ConX	0.9013	0.8278	0.9912	0.9501
	RepLK	0.9002	0.8257	0.9913	0.9503
	Lark	0.8998	0.8283	0.9920	0.9499
	Ours	0.9016	0.8294	0.9922	0.9211
GVLM-CD	ConX	0.8343	0.7641	0.9828	0.9171
	RepLK	0.8352	0.7639	0.9828	0.9176
	Lark	0.8493	0.7631	0.9842	0.9246
	Ours	0.8524	0.7729	0.9850	0.8562

same with RepIFB. Based on the data presented in Table 4, it is evident that except for a slightly lower PA on the SYSU-CD evaluation set compared to Lark, RepIFB outperforms other models across all metrics on the evaluation sets of other datasets. Through comprehensive comparison, the performance of RepIFB is superior to other LKC blocks.

5. Conclusion

In the field of LCCD, existing DL algorithms struggle to capture a universal feature representation of changing features in complex scenes. This challenge leads to difficulties in acquiring structural change information and feature confusion, ultimately affecting the accuracy of detection results.

To address these issues, this study focuses on the application of RLKC in LCCD, CD-RLKNet is proposed. The core of CD-RLKNet lies in the adoption of RLKC to enhance the model's ability to extract global and local features from remote sensing images. The network structure comprises three modules: STAFM, FAAM and BFIM, responsible for spatial-temporal feature fusion of bi-temporal remote sensing data, auxiliary feature extraction, and fine-grained feature extraction, respectively. Among them, STAFM preliminarily fuses bi-temporal remote sensing images by extracting temporal and spatial feature differences, forming a coarse-grained fused feature map. FAAM extracts land cover features from each temporal phase separately to maintain the independence of temporal features. BFIM utilizes BFE-LKB to extract bi-temporal land cover features with a larger receptive field, ensuring the foreground relevance and background independence of detection targets. Experimental results demonstrate that CD-RLKNet achieves the highest metrics on three public datasets, fully proving that RLKC can effectively overcome the limitations of existing DL algorithms in LCCD tasks.

However, we also recognize that while RLKC enhances the model's global feature extraction capabilities, it is crucial to consider computational efficiency and model complexity in practical applications. Furthermore, exploring ways to further optimize the model structure and improve its generalization ability across different scenes remains an important direction for future research.

CRedit authorship contribution statement

Junqing Huang: Writing – original draft, Visualization, Methodology, Conceptualization. **Xiaochen Yuan:** Writing – original draft, Methodology, Conceptualization. **Chan-Tong Lam:** Conceptualization. **Wei Ke:** Conceptualization. **Guoheng Huang:** Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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